

## Samsung SL-65A for Military & Defense

Large-envelope CNC lathe with steady-rest support for precision turning of long, heavy structural round parts. This paper covers what the machine does, the military & defense parts we produce with it, the alloys we run, the tolerances and finishes military & defense buyers expect, and the documentation package.



**Big Thicket Express** · 22" OD × 119" L max turning (with steady rest) · large-envelope heavy-duty CNC lathe with steady rest

### Executive summary

**Machine:** Samsung SL-65A/3200 (shop nickname: *Big Thicket Express*). large-envelope heavy-duty CNC lathe with steady rest at B&R Productions, running from our New Waverly, Texas shop under an ISO 9001:2015 quality system.

**Application:** Large-envelope CNC lathe with steady-rest support for precision turning of long, heavy structural round parts for Military & Defense customers — Defense primes, subassembly manufacturers, tactical vehicle component OEMs, munitions component suppliers, weapon-system integrators, sustainment (out-of-production replacement) contractors.

**Envelope:** 22" OD × 119" L max turning (with steady rest). **Tolerance:**  $\pm 0.0005$ " routine on critical features.

**Traceability:** Material by heat and lot on every job.

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## The machine, in context

This is the heaviest machine on the floor. When a customer sends a 20-inch forged blank and wants it turned to a finished OD with concentric internal features across ten feet of length, Big Thicket Express is the answer. The included steady rest is what makes long slender work possible — without it, deflection eats your tolerance.

### Full spec table

|                             |                                                                                |
|-----------------------------|--------------------------------------------------------------------------------|
| Model                       | Samsung SL-65A/3200 ("Big Thicket Express")                                    |
| Type                        | Large-envelope heavy-duty CNC turning center — Samsung SL-65A platform         |
| Max turning capacity        | 22" OD × 119" L (B&R working spec, with steady rest)                           |
| Platform max swing          | 40.5" over bed                                                                 |
| Platform max turning length | 126" (3,200 mm) between centers                                                |
| Chuck                       | 24" hydraulic 3-jaw                                                            |
| Steady rest                 | Included — supports long slender work to hold straightness                     |
| Main motor                  | 60 HP high-torque                                                              |
| Rigidity                    | Samsung SL-65 platform — heavy cast base for the largest oilfield turning work |
| Tolerance capability        | ±0.0005" routine on critical features; straightness held with steady-rest      |
| Materials                   | Full range — forgings, bar stock, tubing in oilfield and structural alloys     |

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## Who this serves in Military & Defense

**Typical buyer:** Defense primes, subassembly manufacturers, tactical vehicle component OEMs, munitions component suppliers, weapon-system integrators, sustainment (out-of-production replacement) contractors.

**What's hard about military & defense work:** ITAR compliance, security-sensitive prints, program-specific quality plans, tight-tolerance structural components, exotic-alloy selection driven by service environment, and legacy-part sustainment where the OEM no longer supports the design.

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## Typical Military & Defense parts we produce on Big Thicket Express

- Largest-diameter defense turned work
- Long forged structural shafts
- Large actuator and cylinder bodies
- Heavy vehicle-scale structural components
- Extended defense-platform structural rods

Above list is representative; if your part isn't shown, call and ask — most oilfield / aerospace / defense / industrial turned or milled work fits somewhere in the shop.

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## Alloys we run for Military & Defense

**Standard range:** 17-4 PH · 15-5 PH · 13-8 Mo · 4140 pre-hard · 4340 · Titanium Grade 5 · Inconel 718 · Aluminum 7075 · High-strength steels · Customer-specific alloys on request.

Defense parts routinely spec 4140 pre-hard, 4340, and precipitation-hardening stainless — condition-specific machining knowledge matters, same as aerospace. Titanium is common on weight-critical structural components. Legacy sustainment often specifies materials nobody sees anywhere else; we work with the spec sheet you provide and validate machining strategy before we cut.

**What "we run these weekly" means:** we stock or reliably source the common alloys in the industry's typical spec range. Sharp tooling, proven feeds and speeds, and process control specific to each alloy family. Your part is not the one we're experimenting on.

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## **Tolerance and finish expectations in Military & Defense**

±0.0005" routine on critical features. Tighter with proper setup and fixture. GD&T-driven tolerancing supported. CMM verification on first articles; customer-plan-driven measurement on production runs. Documentation the level a program office expects.

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## **Documentation and quality workflow**

ITAR-controlled work under NDA with customer-specific document control — prints stay in-shop. Customer-specific quality plans, PPAP, and first-article documentation standard. Material traceability by heat and lot on every job. Certificates of conformance with delivery.

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## **Response time and emergency work**

Program-driven schedules honored. Emergency component replacement supported for sustainment programs with established relationships. Legacy-part reverse-engineering supported for out-of-production sustainment.

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## **How we compare**

Compared to a large defense-prime captive shop: we're faster for one-offs, small runs, and legacy sustainment where the prime's supply chain has moved on. Compared to a lowest-bidder commodity shop: we handle ITAR, we document properly, and we understand that a program-office audit is a real thing.

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## **Working with B&R Productions**

New customer or existing, the process starts the same way: call (936) 291-7827 or send a print through the quote form. Prints are accepted as PDF, STEP, IGES, DWG, or DXF; if you only have a sample part, we reverse-engineer routinely. NDAs on request. ITAR-controlled prints handled under document control with prints staying in-shop.

Quotes typically come back within 24 hours for standard work, same-day for repeat customers on stocked materials, and near-immediately by phone for emergency and rig-down situations. Lead times are quoted honestly — we don't promise Wednesday and deliver next month.

Ready to talk about your Military & Defense work?

Send us a print or a sample photo and specs. Emergencies: call direct and say "rig-down."

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